

# Henry (Ching Yao) Chen

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## Education

### University of Michigan

Ann Arbor, MI

*Bachelor of Science, Double Major in Data Science and Economics*

May 2026

- Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics, Statistical Computing, Probability, Business Information Systems, Intro to Data Science, Intro to Programming, Linear Algebra, Multivariable Calculus
- Activities: Michigan Hackers, Michigan Undergraduate Students of Statistics, Sigma Eta Pi (Professional Entrepreneurship Fraternity)

## Professional Experience

### Data Science Research Intern

June 2024 – August 2024

*University of Michigan*

Ann Arbor, MI

- Cleaned and transformed a dataset of over a million Android APK files, converting JSON raw data files into master tables
- Streamlined the EDA process for the dataset using Pandas and Seaborn to visualize key features with over 5 graphs
- Developed and implemented an XGBoost model with PySpark, achieving 80% accuracy in identifying malicious apps
- Accelerated virus detection process for APK files by building API call pipelines, extracting, and compiling security reports
- Collaborating with faculty and research leads to enhanced understanding of factors leading to the removal of Android applications from Google Play, reducing processing time by 30%

### Data Analyst

September 2023 – Present

*Michigan Data Science Team*

Ann Arbor, MI

- Actively involved in a project team that builds AI-driven applications and learning the basic concepts of LLM agents
- Demonstrated proficiency in data manipulation and visualization using NumPy, Matplotlib, Seaborn, and Pandas, creating over 10 comprehensive visualizations
- Optimized and cleaned a consumer dataset with 12,000 records using SQL, improving data quality by 20%
- Utilized PowerBI to analyze data, design visualizations, and present results via 5 dashboards and pivot tables
- Built a KNN machine learning model and implemented linear regression, achieving an 80% fit

## Projects

### Economic Trends Forecasting | Python, Scikit-learn, Pandas, Matplotlib

September 2024 – November 2024

- Developed an economic forecasting model using NumPy and Scikit-learn to analyze macroeconomic indicators such as interest rates, inflation, GDP, and unemployment, leveraging time series forecasting techniques
- Performed exploratory data analysis and statistical modeling using visualization libraries (Matplotlib, Plotly) and hypothesis testing to identify key economic trends and insights
- Integrated and cleaned multiple financial datasets using data wrangling techniques, optimized feature engineering, and applied regression analysis to enhance the accuracy of forecasting models

### Analyzing Students' Mental Health | Python, SQL, Pandas, Matplotlib

September 2023 – November 2023

- Generated and interpreted SQL queries to analyze mental health data of international students using MySQL, investigating the relationship between length of stay, social connectedness, acculturative stress, and depression levels
- Visualized the dataset to show collaboration using Pandas and Matplotlib, creating 5 interactive graphs

### NFL Play-by-play Modeling and Classification | Python, Pandas, Matplotlib, Seaborn, NumPy

November 2023

- Cleaned and aggregated a dataset of 50,000 rows to find confidence intervals, improving data integrity by 25%
- Visualized a 50,000-row dataset using Matplotlib and Seaborn with 10 scatterplots and boxplots
- Performed linear regression of the outdoor and indoor games, conducting hypothesis testing with a 95% confidence level

## Technical Skills

**Languages:** C/C++, Python, SQL (Postgres, MySQL), R Studio

**Analytics Tools:** PowerBI, Tableau, Advanced Microsoft Excel

**Developer Tools:** VS Code, Visual Studio, Git

**Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PySpark

**Certifications:** Associate Data Analyst Certification (DataCamp)